

Vector Database and RAG

Vector Store

Architecture

In the previous sessions, you learnt in detail about vector embeddings and their representations for various types of unstructured data, including textual data. And in the previous session, we built a semantic search application and stored the vector embeddings locally using a Pandas dataframe.

While easy, retrieving and storing vector embeddings locally, such as in a dataframe, is quite time-consuming when the operation is scaled to include multiple documents. This can give rise to latency issues in our semantic search application, especially when real-time search and retrieval are required. To cater to such low-latency requirements, developers are increasingly using vector stores to store vector embeddings once documents are ingested and chunked. In the video below, Aditya will explain the various aspects of vector stores.

upGrad

Features	Relational Database	Vector Store
Architecture 	Traditional relational databases are designed for storing and querying structured data	Vector stores are specifically designed for storing unstructured data, such as text or images in the form of vector embeddings
Data Storage 	Traditional relational databases store data in tables with rows and columns	Vector embeddings are stored as vectors or arrays
Querying 	Querying traditional databases involves comparing specific values in specific columns.	Vector stores are optimized for similarity search, clustering, and classification tasks
Information Retrieval 	Information retrieval of vector embeddings from a traditional RDBMS is sub-optimal	Information retrieval of vector embeddings is efficient due to vector-specific indexing methods like HNSW or IVF that can retrieve similar vectors quickly

As mentioned in the video, a vector store is a data storage system that is specially designed to store, index and retrieve high-dimensional vectors quickly. They can store and retrieve vectors much faster than traditional relational databases (RDBMS) and local storage options such as dataframes or flat files (.csv, .xlsx etc.). The image below illustrates some of the key differences between relational databases and vector stores.

Features	Relational Database	Vector Store
Architecture 	Traditional relational databases are designed for storing and querying structured data	Vector stores are specifically designed for storing unstructured data, such as text or images in the form of vector embeddings
Data Storage 	Traditional relational databases store data in tables with rows and columns	Vector embeddings are stored as vectors or arrays
Querying 	Querying traditional databases involves comparing specific values in specific columns.	Vector stores are optimized for similarity search, clustering, and classification tasks
Information Retrieval 	Information retrieval of vector embeddings from a traditional RDBMS is sub-optimal	Information retrieval of vector embeddings is efficient due to vector-specific indexing methods like HNSW or IVF that can retrieve similar vectors quickly

They offer fast and accurate similarity search and data retrieval based on their vector distance or other similarity metrics. Vector stores consist of various components that work together to provide efficient storage, indexing and querying capabilities for high-dimensional vectors. Some of its key features include data management, metadata storage and filtering, and approximate nearest-neighbour (ANN) search algorithms.

- **Data management:** Vector databases offer features for easy data storage, insertion, deletion and updating, making it convenient to manage and maintain vector data.
- **Metadata storage and filtering:** These databases can store metadata associated with each vector entry, allowing users to query the database using additional metadata filters for more precise queries.
- **Approximate nearest-neighbour (ANN) search algorithms:** Vector databases use a combination of algorithms to optimise similarity search, such as hashing, quantisation or graph-based search.

The main advantages of using vector stores for storing and querying high-dimensional vectors are as follows:

- **Fast and accurate similarity search:** Vector stores excel at finding the most similar or relevant data based on the underlying semantic or contextual meaning of various texts which enables efficient retrieval of information. These can return query results faster than the traditional methods of search, such as keyword-based search or k-Nearest Neighbour-based searching methods.
- **Flexibility:** Vector stores can be used with various types of high-dimensional vectors, ranging from tens to thousands of dimensions, depending on the complexity and granularity of the data.

The popularity of vector stores is augmented by the availability of indexing strategies that can retrieve embeddings faster than traditional lookup-based approaches. Indexing in vector stores involves breaking down a document or website into smaller segments and converting these segments into vectors that can be stored in a vector database. The indexing process maps the vectors to a data structure that can be traversed quickly.

In the previous segment, you were introduced to the two types of vector stores: vector libraries and vector databases. In the video below, Aditya will explain these terms in detail.

Indexing:

As mentioned in the video, vector stores use indexing strategies to efficiently query vectors by computing the proximity of a query to the vector embeddings. The indexing algorithms used in

vector databases vary depending on the specific application. Recently, however, approximate nearest-neighbours (ANN) methods such as product quantisation, Hierarchical Navigable Small World (HNSW) and Locative Sensitive Hashing (LSH) have garnered significant attention from developers and researchers alike. As the name suggests, ANN methods involve an approximation of the usual nearest-neighbour methods. You might already be familiar with some of the common methods of the nearest-neighbour algorithm called the k-nearest neighbour (kNN) method.

As discussed already, semantic search involves comparing the vector representations of a query and document by generating the vector embeddings and comparing the embeddings using a distance metric such as cosine similarity. Exact nearest neighbours, such as the kNN algorithm, can often help narrow down the retrieval process and produce accurate search results. But this accuracy comes at the expense of increased retrieval time. On the other hand, ANN methods sacrifice accuracy for speed.

ANN-> Same as Euclidean in L2 and $1/n$ th in n dimensional space
Euclidean Distance- $\sqrt{(x-y)^2}$

$$\sqrt{((1-0)^2+(0-1)^2)}=1.414$$

Now, as mentioned in the previous session, such an exact search often results in an $O(N)$ time complexity; however, ANN techniques result in a sub-linear time complexity $O(\log(N))$. This is achieved with the help of special indexing techniques that make retrieval faster compared to traditional lookup-based methods. Indexing is like sorting a guest list by a certain characteristic, such as the first letter of their names or their closeness to you, so you can find your friends faster. Searching in vector stores involves querying the vector database to retrieve the most similar or relevant data based on their vector distance or similarity. The vector store compares the indexed query vector to the indexed vectors in a data set to find the nearest neighbours by applying a similarity metric of the indexed vectors.

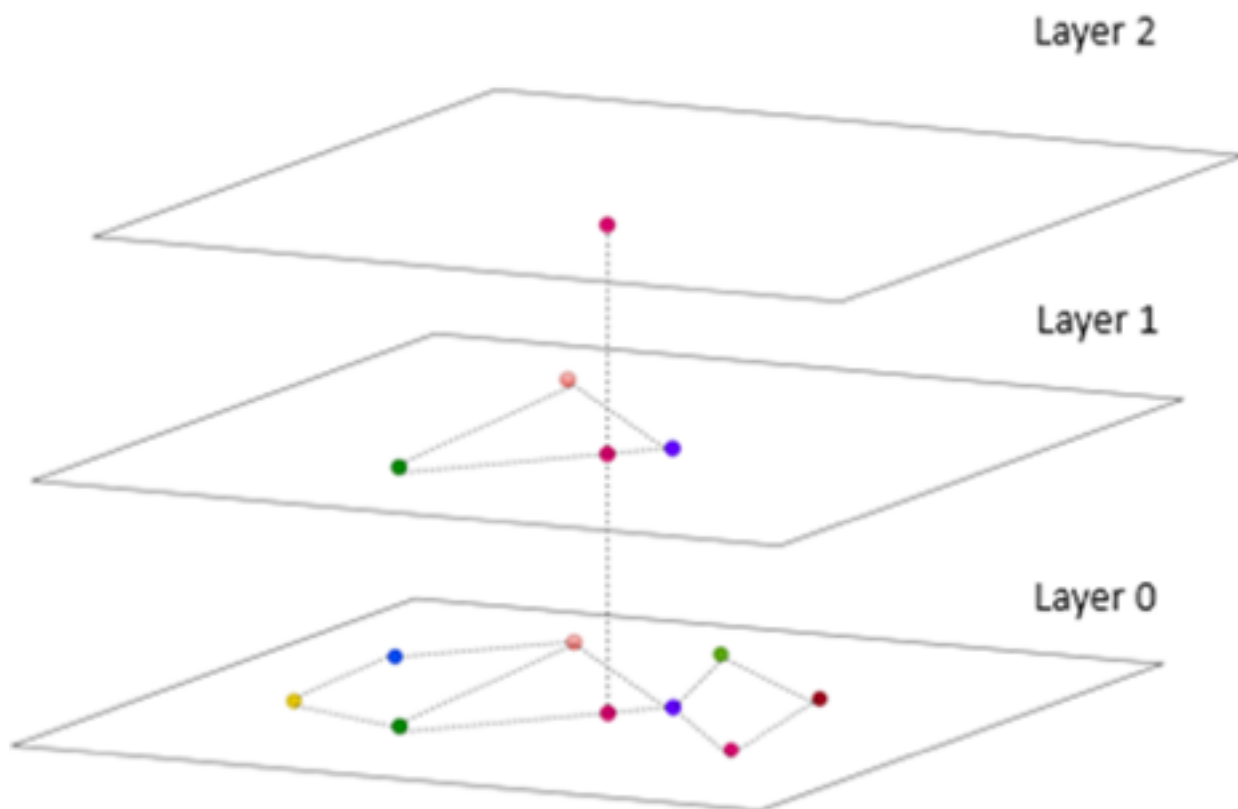
In summary, indexing is the process of organising vectors in a way that allows for efficient similarity search, while searching is the process of querying a vector database to retrieve the most similar or relevant data based on their vector distance or similarity. Some of the common approximate nearest-neighbour algorithms are as follows:

- Tree-based algorithms such as ANNOY, which was created by Spotify
- Graph-based algorithms such as the Hierarchical Navigable Small World (HNSW) algorithm; popular C++ implementation of this algorithm available [here](#)
- Cluster-based algorithms such as the [Facebook AI Similarity Search \(FAISS\)](#) and [Product Quantisation \(PQ\)](#)
- Hash-based algorithms such as [Locality Sensitive Hashing \(LSH\)](#).

Each algorithm mentioned above finds applications in various use cases and comes with its own advantages and disadvantages. In the video above, Aditya explained the popular algorithm HNSW, which is a popular method of conducting approximate nearest-neighbour searches.

Hierarchical Navigable Small World (HNSW)

The Hierarchical Navigable Small World (HNSW) algorithm is a popular graph-based method that combines the principles of Navigable Small World and proximity graphs. It is a fully graph-based solution that constructs a multi-layered graph with fewer connections in the top layers and more dense regions in the bottom layers as shown in the image below. The search starts from the highest layer and moves one level below every time the local nearest neighbour is found greedily among the layer nodes. Ultimately, the nearest neighbour found in the lowest layer is the answer to the query. Nodes in HNSW are inserted sequentially one by one, and every node is randomly assigned an integer indicating the maximum layer at which the node can be present in the graph.



The HNSW greedy search algorithm is sublinear, which means it has a complexity close to $\log(N)$, where N is the number of vectors in the graph. This makes it an efficient algorithm for approximate nearest-neighbour search. HNSW is used in various vector databases and libraries, including Pinecone, Faiss and ChromaDB.

In the next segment, we will go over the various types of vector stores that are commonly available, namely vector libraries and vector databases.

Supplementary Resources:

- This [article](#) provides an in-depth analysis of various vector indexing algorithms.
- This [article](#) by Pinecone elucidates the intricacies of the HNSW algorithm in detail.
- This [article](#) illustrates the BM25 indexing strategy employed in information retrieval.
- In this [article](#), the author delves into the utilization of the Inverted File Index (IVF) indexing strategy.

Vector Library

Vector stores come in multiple flavours; two common categories are vector libraries and vector databases.

Vector libraries and vector databases can be used to efficiently perform nearest-neighbour searches to retrieve similar pieces of text based on their semantic meaning. Vector libraries (also referred to as vector indices or vector search libraries) are popular for quick prototyping purposes and when the data size is considerably small. These libraries do not support the usual create, read, update and delete (CRUD) support that traditional relational databases and vector databases offer; hence, they are not suitable for building scalable applications. They, however, offer native support to store the vector embeddings to the local disk by persisting it from memory to the local disk. Popular vector libraries include [Meta's FAISS](#), [Spotify's ANNOY](#), [Google ScaNN](#), [NMSLIB](#), [hnswlib](#), etc.

Vector databases are optimised for storage and the retrieval of vector embeddings with the additional capabilities to store and update the vector embeddings, as they support CRUD operations natively. This makes them a great choice for applications that require low-latency search, such as recommendation systems, search engines and chatbots. Vector databases are typically more focussed on enterprise-level production deployments as opposed to vector libraries that are used for quick prototyping. Popular vector databases include [Chroma](#), [Pinecone](#), [Weviate](#), [Qdrant](#), etc.

The figure below illustrates the differences between both types of vector stores.

Feature	Vector Libraries	Vector Databases
 Storage	Can be persisted to disk	Native support to store on disk
 Optimization	Features for optimization available	Has complete access to features for optimization
 Querying	Performance optimized for in-memory similarity search	Can perform search, object retrieval from persistent storage, optional inverted index filtering, network requests
 Updatability (CRUD)	No	Yes
 Speed	Faster than traditional RDBMs	Slower, in comparison to vector libraries
 Durability, Crash Recovery	No	Yes

Additionally, vector databases solve several major limitations of vector libraries. These include:

Feature	Vector Libraries	Vector Databases
Storage	Can be persisted to disk	Native support to store on disk
	Optimization Features for optimization available	Has complete access to features for optimization
Querying	Performance optimized for in-memory similarity search	Can perform search, object retrieval from persistent storage, optional inverted index filtering, network requests
Updatability (CRUD)	No	Yes
Speed	Faster than traditional RBDMS	Slower, in comparison to vector libraries
Durability, Crash Recovery	No	Yes



Index is stored in library..

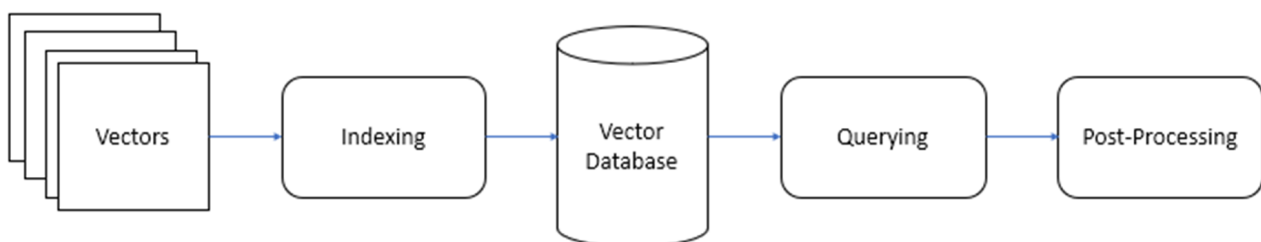
Additionally, vector databases solve several major limitations of vector libraries. These include:

- **Metadata storage and filtering:** Vector databases can store metadata associated with each vector entry. Users can then query a database using additional metadata filters for finer-grained queries. Vector libraries store only vector embeddings and not the associated objects they were generated from. When you run a query, a vector library will respond with the relevant vectors and object IDs. This is limiting since the actual information is stored in the object and not the ID. To solve this problem, you would have to store the objects in a secondary storage. You could then use the returned IDs from the query and match them to the objects to understand the results.
- **Scalability:** Vector databases are designed to scale with growing data volumes and user demands, providing better support for distributed and parallel processing. Standalone vector indices may require custom solutions to achieve similar levels of scalability (such as deploying and managing them on Kubernetes clusters or other similar systems).
- **Real-time updates:** Vector databases often support real-time data updates, allowing for dynamic changes to the data, whereas standalone vector indexes may require a full re-indexing process to incorporate new data, which can be time-consuming and computationally expensive.
- **Backups and collections:** Vector databases handle the routine operation of backing up all the data stored in a database. Pinecone also allows users to choose specific indexes that can be backed up in the form of 'collections', which store the data in that index for later use.
- **Ecosystem integration:** Vector databases can more easily integrate with other components of a data processing ecosystem, such as ETL pipelines (like Spark), analytics tools (like Tableau and Segment) and visualisation platforms (like Grafana), streamlining the data management workflow. It also enables easy integration with other AI-related tools such as LangChain, LlamaIndex and ChatGPT's plugins.
- **Data security and access control:** Vector databases typically offer built-in data security features and access control mechanisms to protect sensitive information, which may not be available in standalone vector index solutions.

Here are the steps in the typical process of storing vector embeddings in a vector database:

1. **Generate vector embeddings:** In this step, you generate the vector embeddings for the documents.
2. **Perform indexing:** The vector database indexes vectors using an algorithm such as PQ, LSH or HNSW (more on these below). This step maps the vectors to a data structure that will enable faster searching.
3. **Store indices and embedding vectors:** In this step, you store the embedding vectors in the local storage or cache of the vector store and generate the indices for the embeddings.
4. **Querying:** The vector database compares the indexed query vector to the indexed vectors to find the nearest neighbours (applying a similarity metric used by that index)
5. **Perform post-processing:** In this stage, common data management techniques such as updation and deletion operations are performed on the vector embeddings. In some cases, the vector database retrieves the final nearest neighbours from the data set and post-processes them to return the final results. This step can include re-ranking the nearest neighbours using a different similarity measure.

LL



Generative Search

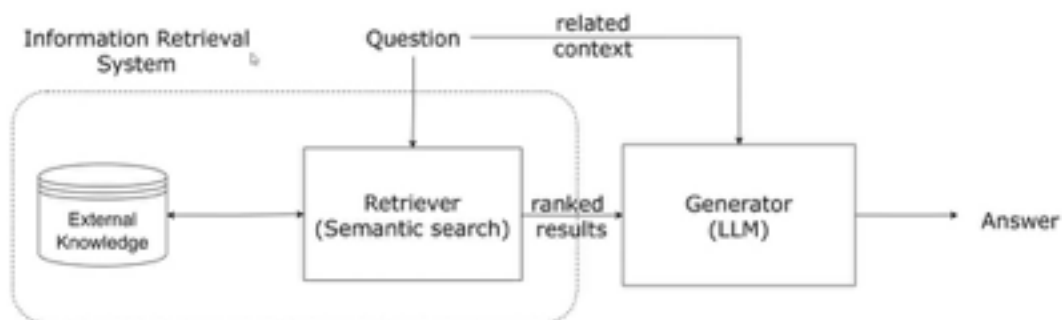
Overview

LLMs have their knowledge of their own. We as a company have our own database. We need to first decide what data should be public to LLM and private to us. Using this public data, we tell the LLM to learn from this data, provided as embeddings as tensors and this and knowledge available to it on its own and web search, it can automate work of our own. Like Private cloud compute. Our data is stored there, which is encrypted even for Apple, but chatgpt can store this data in its servers when Siri is using this plugin to answer our queries. What's the point of this then? Microsoft is known for its violation of privacy easily when we use other services which almost every software is using chatgpt to answer. There has to be a better way. Yes Siri Plus gemini apparently. Do not know the architecture.

In the previous session, you were introduced to the key elements of semantic search. You dived into the technicalities of the semantic search pipeline and also learned how to augment the performance of the semantic search pipeline with the help of vector databases. Semantic search is a type of search that goes beyond traditional keyword matching to understand the meaning of a query and return results that are relevant to the user's intent. You were also introduced to the term 'generative search', which refers to the type of search that uses artificial intelligence to generate new content in response to a user's query. This content can include text, images, code and creative content. Let's hear more about it in the video given below.

upGrad

Retrieval Augmented Generation (RAG)



As explained in the video, the main distinction between semantic search and generative search is that semantic search is primarily focused on retrieving relevant information, whereas generative search is focused on generating new content. However, the two technologies can be used

together to improve the performance of a variety of tasks, such as question answering, summarisation, and machine translation. For example, a question-answering system can use semantic search to retrieve relevant documents from a knowledge base and then use generative search to generate a comprehensive and informative answer to the user's question. The differences between semantic search and generative search are elaborated in the table below.

Features	Semantic Search	Generative Search
Information Retrieval 	Semantic search results are limited to retrieving relevant documents based on the query and then extracting answers from those documents.	Generative search employ generative models to directly generate the responses to queries.
Natural Language Response 	The semantic search responses are limited in their responses as they only extract the relevant documents.	They can retrieve the relevant information and can also generate answers in natural language offering a conversational output.

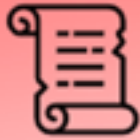
Overall, semantic search and generative search are two powerful technologies that can be used to improve the performance of various AI tasks. Using these technologies together, we can create AI systems that are more accurate, informative and helpful.

Retrieval augmented generation (RAG)

Retrieval augmented generation (RAG) is a special type of generative search that combines the strengths of semantic search and large language models to generate more accurate responses to user queries. This is a new search paradigm that combines the strengths of both retrieval-based models and generative foundation models to enhance the quality and relevance of the generated text. RAG retrieves relevant information from an external knowledge base to supplement the LLM's internal representation of information, which allows for fine-tuning and adjustments to the LLM's internal knowledge, making it more accurate and up-to-date. RAG has several applications, including question-answering systems, chatbots, and industry-specific LLMs. RAG can reduce hallucinations and repetition while improving specificity and factual grounding compared with conversation without retrieval. RAG can also provide more contextually appropriate answers to prompts as well as base those answers on the latest data.

RAG Pipeline

- Embedding Layer
- Search and Rank Layer
- Generation Layer



Embedding Layer



Search & Rank Layer



Generation Layer

Now, let's discuss each of these layers in detail.

Embedding Layer

You are already familiar with the embedding layer, as it was covered in the previous sessions on semantic search. The embedding layer is typically the first layer of a RAG model, and it typically contains an embedding model that is trained on a massive data set of text and code. This data set is used to learn the relationships between words and phrases and to create embeddings that represent these relationships. The embedding layer is an important part of RAG models because it allows your system to understand the meaning of the text that it is processing and understand its semantic relationship to the query. The embedding layer generates embeddings for your text corpus and allows the RAG model to understand the meaning of the query and to generate a relevant and informative response. This is essential for a variety of tasks, such as question answering, summarisation and machine translation.

Search and Rank Layer

The next layer is the search and rank or the re-rank layer. The search and re-rank layer is a crucial component that is responsible for retrieving the relevant information from an external knowledge base, ranking it based on its relevance to the input query and presenting it to the generation layer for further processing. The search and re-rank layer is an essential component of RAG, as it ensures that the retrieved text is accurate, relevant and contextually appropriate. The search and re-rank layer typically consists of two components:

- A search component that uses various techniques to retrieve relevant documents from the knowledge base
- A re-rank component that uses a variety of techniques to re-rank the retrieved documents to produce the most relevant results

The search component typically uses a technique called semantic similarity. As discussed in the previous session, semantic similarity is a measure of how similar two pieces of text are in terms of their meaning. The search component uses semantic similarity to retrieve documents from a knowledge base that are relevant to the user's query.

The re-rank component of the search typically uses a variety of techniques to re-rank the retrieved documents. These techniques can include the following:

- Ranking by relevance: The re-rank component can rank the retrieved documents based on how relevant they are to the user's query.
- Ranking by popularity: The re-rank component can rank the retrieved documents based on how popular they are, such as by measuring the number of times they have been viewed or shared.
- Ranking by freshness: The re-rank component can rank the retrieved documents based on how recent they are, such as by measuring the date on which they were published.

The search and re-rank layer is an important part of RAG models because it allows the model to retrieve and re-rank relevant documents from a knowledge base. This is essential for numerous tasks, such as question answering, summarisation and machine translation. The search and re-rank layer is a powerful tool that can be used to improve the performance of a variety of AI tasks. It is an essential part of RAG models, and it plays a key role in helping these models retrieve and re-rank relevant information. The retrieval-based model is used to find relevant information from existing information sources. The re-rank layer is used to rank the retrieved information based on its relevance to the input query.

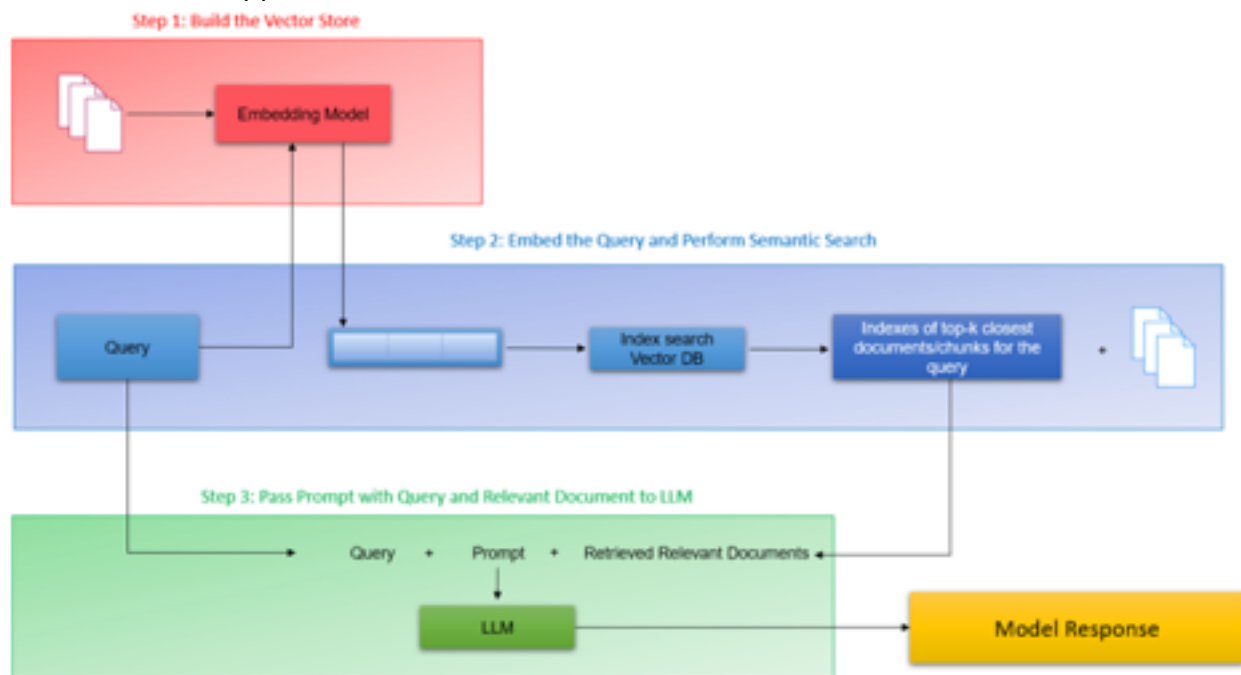
Generation Layer

The generation layer is typically the last layer of a RAG model which consists of a foundation large language model that is trained on a massive data set of text and code. As the name suggests, the generation layer allows the model to generate new text in response to a user's query. The generative model takes the retrieved information, synthesises all the data and shapes it into a coherent and contextually appropriate response. This is essential for many tasks, such as question answering, summarisation machine translation and also generative search specifically RAG. In the context of search, this layer excels in providing context and natural language capabilities for generative search.

Aditya then explained the various stages of the RAG pipeline as shown in the image below. These are given below:

1. **Step 1:** Build the vector store
2. **Step 2:** Embed the query and perform semantic search
3. **Step 3:** Pass the prompt with the query and the relevant documents to a Large Language Model (LLM)

NOTE: In this project, we will be using GPT-3.5 as the Large Language Model for building the generative search application.



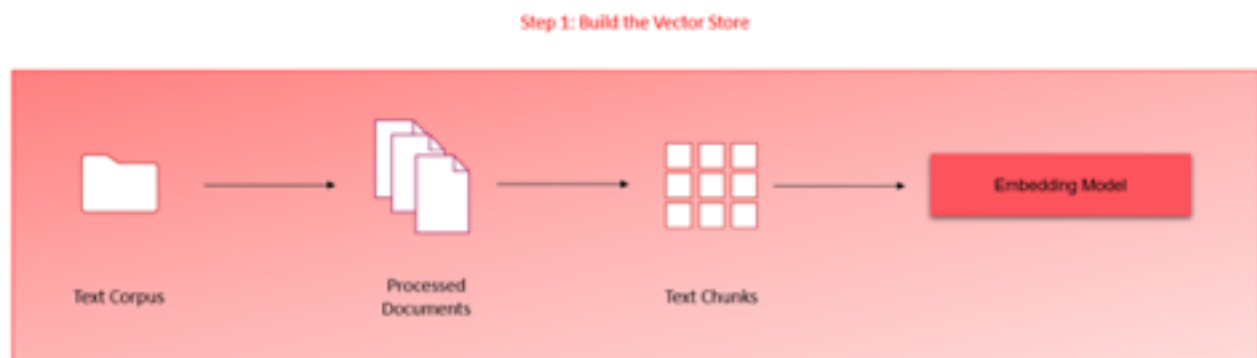
Let's go over each of these steps in detail.

Step 1: Build the vector store: The first step is to build a vector store that can store documents along with metadata. A vector store is a database that stores embeddings of text data in a vector space. The documents are converted to raw text and then split into chunks. Each chunk is then represented as a vector using an embedding model. The vector store is then populated with these vectors.

Step 2: Embed the query and perform semantic search: The next step is to embed the user query into the same vector space as the documents in the vector store. This is done using an embedding model. Once the query is embedded, a semantic search is performed to find the closest embedding from the vector store. The entries with the highest semantic overlap with the query are retrieved.

Step 3: Pass the prompt with the query and the relevant documents to the LLM: The final step is to pass the prompt, which is a concatenation of the query and the retrieved documents, to the LLM. The LLM generates a response based on the context of the query, the system prompt and the relevant documents passed from the search layer. The retrieved documents serve as the knowledge bank and provide the necessary context for the query to the LLM, which helps it generate a more accurate and relevant response.

In the first step, we process the documents (in this case, the documents pertain to the insurance domain) to extract the text, split it into smaller chunks and then pass them to the embedding model.



As you saw in the video, the [PDFPlumber library](#) is very efficient in extracting the text contents of multiple PDF documents. The library can also represent a table in a neat list of lists format that preserves the original hierarchical structure of the document. The library also supports visual debugging of almost any type of machine-generated PDFs; you can read more about it in the [documentation link](#) provided above.